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ARTIFICIAL INTELLIGENCE, DIGITALIZATION, AND NEW TECHNOLOGIES | REVIEW ARTICLE OPEN ACCESS Check for updates

Artificial Intelligence agents and autonomous decision-making in business: a review

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ABSTRACT

Artificial intelligence (AI) agents capable of autonomous decision-making are increasingly transforming business processes and managerial decision structures. Despite growing adoption, research remains fragmented across domains and lacks a unified understanding of how agentic AI is conceptualized, applied, and governed in business contexts. This study conducts a systematic scoping review of literature published between 2020 and 2025 using major academic databases. A total of 875 studies were included for quantitative mapping and thematic analysis, with a representative subset selected for in-depth qualitative synthesis. Findings reveal three dominant conceptual lenses of AI agents (technical, organizational, and hybrid), with applications concentrated in strategy, operations, and human resource management. Reinforcement learning, simulation, and optimization are the most commonly used techniques. While AI agents contribute to efficiency gains and faster decision-making, significant challenges related to accountability, transparency, and system integration remain. We propose an Agentic AI in Business (AAB) integration model and a taxonomy distinguishing decision-support, semi-autonomous, and fully autonomous systems. The study offers conceptual clarity and practical insights for organizations and policymakers regarding governance, trust, and responsible adoption of autonomous AI systems. Policymakers should develop differentiated governance frameworks that calibrate regulatory oversight to the level of AI autonomy, ensuring accountability without impeding innovation.

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Agentic Artificial Intelligence; autonomous decision-making; business management and strategy; AI governance and accountability; scoping review; decision authority

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Artificial Intelligence; Systems & Control Engineering; Computer Science (General); Business, Management and Accounting

1. Introduction

Decision-making is an integral part of managers' daily work, yet it is becoming increasingly complex as organizations must evaluate multiple alternatives, conflicting criteria, uneven expertise, and future (Aziz et al., 2016; Sorooshian, 2021; Sorooshian et al., 2023). In response to this growing complexity, artificial intelligence (AI) agents capable of autonomous decision-making are increasingly transforming business processes and management decisions (Dvouletý et al., 2025; Elgendy et al., 2025). According to a 2023 global survey, 42% of enterprises have actively deployed AI and another 40% are experimenting with AI solutions, reflecting the rapid uptake of AI-driven decision tools in industry (Bonderud, 2025). These agentic AI systems, designed to operate with minimal human intervention, promise to revolutionize business decision-making by combining algorithmic speed and accuracy with vast data–driven insights (Elgendy et al., 2025). In domains including finance and operations, AI agents have been used to support or even replace managerial decisions, yielding improvements in decision agility, efficiency, and consistency. For example, companies adopting AI to assist strategic planning have reported up to 42% faster decision-making and 18% cost savings while maintaining high management satisfaction when the AI's reasoning is transparent (Acuña Acuña, 2025). Such gains illustrate the potential for AI agents to enhance business outcomes and innovation.

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Early evidence from various sectors demonstrates the benefits of autonomous AI decision-makers. In financial services, agentic AI systems are being applied to areas like algorithmic trading, fraud detection, and customer service automation, where they can provide adaptive, context-aware decisions at scale (Elgendy et al., 2025). For example, AI "concierge" chatbots can handle customer inquiries with high accuracy, converting a significant proportion into leads without human intervention (Bonderud, 2025). In operations and supply chain management, multi-agent reinforcement learning (MARL) systems enable decentralized, data-driven coordination, such as adaptive pricing or inventory control, that improves overall efficiency and responsiveness (Zhang et al., 2024). These advances align with reports that two-thirds of business leaders see AI leading to over 25% improvement in revenue growth rates (Bonderud, 2025). Autonomous AI agents have begun to deliver tangible performance gains by augmenting or accelerating business decisions across various functions.

Alongside these opportunities, significant challenges and concerns have emerged regarding AI-driven autonomous decision-making. A recurrent issue is accountability and trust, such as where responsibility lies if an AI agent makes a poor or unethical choice (Acuña Acuña, 2025; Ghosh, 2025). Both business practitioners and researchers are concerned about the "black box" nature of many AI agents, as algorithmic opacity can undermine users' trust and hinder adoption (Hughes et al., 2025). Indeed, even when AI agents perform well, stakeholders remain cautious if the decision logic is not transparent or the stakes are high (e.g. life-and-death decisions in healthcare) (Bonderud, 2025). Other critical challenges include bias in AI-driven processes, integration difficulties with legacy systems, and the need to ensure that AI decisions comply with regulations and societal values (Hughes et al., 2025). In summary, the rise of autonomous decision-making agents in business has outpaced human frameworks for governance, ethical oversight, and organizational accountability. Nevertheless, AI agents represent a paradigm shift in which autonomous agents work either individually or collaboratively to address decision challenges (Piccialli et al., 2025).

The increasing autonomy of AI agents introduces fundamental questions regarding how decision authority is distributed and governed and how AI judgment is trusted within organizations. As AI systems move from decision-support tools toward semi-autonomous and fully autonomous agents, traditional managerial roles, accountability structures, and governance mechanisms are being reshaped. This shift calls for a systematic understanding of how autonomous decision-making is conceptualized and implemented across business contexts.

A growing body of literature reviews has examined AI in business and management. For example, Bankins et al. (2024) provide a multilevel review of AI in organizations, emphasizing implications for organizational behavior such as human–AI collaboration, worker attitudes, and algorithmic management. Ramaul et al. (2026) problematize how AI is theorized in management research, showing that current perspectives often reduce AI to either a perfectly rational actor or a human-like counterpart, thereby overlooking its distinct role in organizational decision-making structures. Numerous reviews have examined AI use in operations and production. Antons and Arlinghaus (2022) contrast centralized and decentralized decision authority in the "factory of the future," while Gerrits et al. (2024) synthesize self-organizing logistics and distributed agent-based control. George et al. (2025) integrate fragmented findings on AI in frontline services (trust and acceptance), and Sidlauskiene (2022) discusses drivers of consumer adoption of intelligent agents. Stylos et al. (2025) propose a framework for human–AI collaboration and legitimacy in hybrid decision settings in tourism.

Table 1 summarizes these representative reviews. Collectively, they demonstrate breadth but little cohesion across sectors and lenses. Crucially, no prior review has systematically examined how AI-driven autonomy reshapes decision-making structures across business functions. Existing reviews remain fragmented, focusing on either specific domains (e.g. operations, marketing, logistics) or adjacent issues such as consumer adoption and workforce attitudes, without directly addressing how decision authority, processes, and governance mechanisms are being transformed by agentic AI. Although governance, transparency, and human–agent collaboration appear in some reviews (e.g., Stylos et al., 2025), there has been no discussion of cross-domain integration or how decision authority and processes are being reconfigured by agentic AI. This gap motivates the present scoping review, which synthesizes the decision-centered implications of AI agents across business contexts.

Table 1. Representative reviews of AI in business contexts.

Study	Focus domain/scope	Main contribution	Limitation/gap relative to this study
Bankins et al. (2024)	AI in organizations/organizational behavior	Multilevel review: collaboration, algorithmic management, attitudes	Emphasizes workforce outcomes, not decision structures per se
Ramaul et al. (2026)	Management theory	Problematizing review of AI theorization (rational vs. anthropomorphic views)	Lacks actionable frameworks for decision authority reconfiguration
Antons and Arlinghaus, (2022)	Manufacturing/Industry 4.0	Centralized vs. decentralized decision authority in factories	Domain-specific; limited cross-functional generalization
Gerrits et al. (2024)	Logistics and transportation	Self-organizing logistics; typology for distributed control	Conceptual, sector-specific; no cross-business synthesis
George et al. (2025)	Frontline services/marketing	Systematic review on trust and acceptance of service AI	Consumer-facing only; limited link to internal decision structures
Sidlauskiene, (2022)	Digital commerce/ consumer adoption	Drivers of adoption of intelligent agents	Focus on consumers, not managerial decision-making
Stylos et al., (2025)	Tourism/hybrid human–AI decisions	Framework for human–AI teaming and legitimacy	Sector-specific; lacks cross-industry integration

Authors' own elaboration.

The reviewed literature has several critical limitations. First, existing studies tend to analyze AI applications within isolated domains such as operations, marketing, or logistics, without providing an integrated cross-functional perspective (Bankins et al., 2024; Stylos et al., 2025). Second, the literature emphasizes technological capabilities and performance outcomes while offering limited insight into how AI-driven autonomy reshapes decision authority, governance structures, and organizational processes (Ramaul et al., 2026). Third, reviews often focus on adoption, ethics, or human–AI interaction but do not systematically differentiate between levels of autonomy or examine their implications for decision-making structures (Bankins et al., 2024; Stylos et al., 2025). These limitations indicate the absence of a structured framework that connects conceptual definitions of AI agents with their roles in business decision-making. Addressing these gaps provides the foundation for the research questions (RQs) guiding this study.

Building on these limitations, there is a pressing need for a structured and integrative synthesis that focuses on how AI autonomy restructures decision rights, processes, and governance across business functions and on what frameworks can guide responsible adoption at different autonomy levels. This study is motivated by the need to reduce this fragmentation and provide a structured, cross-functional understanding of autonomous decision-making in business. By placing decision structures at the center of analysis, this review offers an integrative perspective that connects conceptual definitions, application domains, technological foundations, and governance considerations.

The significance of this study lies in its ability to support both scholarship and practice. For researchers, it provides conceptual clarity and a foundation for theory development in the emerging field of agentic AI. For practitioners and policymakers, it offers insights into how different levels of autonomy can be responsibly implemented, governed, and aligned with organizational objectives.

As AI continues to transform business processes, a comprehensive understanding of how autonomous agents are being integrated and what implications they carry is essential. Thus, this article aims to provide a timely and rigorous synthesis of recent advances, offering a foundation for both theoretical development and practical deployment of AI in organizational decision-making. By systematically surveying the literature from 2020 to 2025, it reveals the current state of the art, highlights emerging trends, and identifies critical gaps that warrant further investigation. In doing so, the review not only maps out where and how AI-driven agents are influencing business outcomes but also underscores the challenges that must be addressed to harness their full potential responsibly.

This review is theoretically informed by several established frameworks in business and management scholarship. The allocation of decision rights between humans and AI agents is increasingly theorized as a dynamic process of cognitive reapportionment, wherein decision authority is redistributed across human and system co-cognitors based on the nature of the decision (Konsynski et al., 2024). As AI systems evolve toward greater autonomy, agency theory provides a lens for understanding how decision rights and control mechanisms should be reconfigured to align AI agents with organizational interests

(Humberd & Latham, 2026). Socio-technical systems thinking frames this process as a reconfiguration of workflows, accountability boundaries, and professional roles, producing both efficiency gains and governance tensions (de Gennaro et al., 2026). Complementing these perspectives, algorithmic management research examines how AI systems increasingly govern managerial tasks and work processes within organizations, raising fundamental questions about human oversight, trust, and accountability (Hillebrand et al., 2025; Raman et al., 2026). Together, these theoretical perspectives position the review within contemporary management scholarship and provide a conceptual foundation for interpreting the findings.

The remainder of this paper is organized as follows. Section 2 outlines the scoping review methodology, including the literature search strategy, selection criteria, and analytical approach. Section 3 presents the key findings of the review, structured around the major themes emerging from the RQs. These themes include conceptual definitions of AI agents; their application domains in business; the technical approaches underpinning their autonomy; reported outcomes and impacts; identified challenges and risks; and the frameworks or models proposed for governing their use, along with suggested future directions. Section 4 provides a critical discussion of these findings, identifying implications for both research and practice and highlighting remaining knowledge gaps. Section 5 concludes the paper by summarizing the contributions of the review and offering recommendations for future research aimed at advancing the effective and responsible use of autonomous AI agents in business.

2. Methodological overview

We adopted a systematic scoping review approach to map and synthesize the emerging literature on AI agents and autonomous decision-making in business contexts. A scoping review is particularly appropriate for this study due to the interdisciplinary, rapidly evolving, and conceptually fragmented nature of the field, in which definitions, applications, and theoretical perspectives are not yet standardized (Arksey & O'Malley, 2005; Peters et al., 2015). Unlike traditional systematic reviews, which focus on narrowly defined questions and effect size aggregation, scoping reviews are designed to provide broad coverage of complex bodies of literature, identify research gaps, clarify key concepts, and map emerging themes across diverse domains (Peters et al., 2015; Riandhi et al., 2025). This approach is therefore well-suited to capturing the multidimensional nature of agentic AI in business, which spans technical systems, organizational processes, and governance considerations. In addition, this study combines elements of scoping review and bibliometric mapping to enhance analytical rigor. While the scoping framework enables comprehensive coverage and conceptual synthesis, bibliometric techniques support the identification of patterns, trends, and thematic structures across a large dataset. This combined approach allows both breadth and depth in analyzing AI-driven decision-making in business.

We used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, and Figure 1 shows the PRISMA flowchart. Figure 2 illustrates the methodological steps followed in this review. These steps were designed to ensure rigor, transparency, and reproducibility while capturing the wide variety of perspectives represented in the literature.

Following the filtering process, 1,012 records were retained. After removing 137 duplicate records, a total of 875 unique articles were included in the final dataset. These records formed the basis for quantitative mapping and thematic analysis. A smaller subset of representative studies was selected for in-depth qualitative discussion.

A. Defining Objectives and Research Questions

We established six guiding RQs (listed below) to ensure comprehensive coverage of conceptual, empirical, and practical aspects of agentic AI in business.

B. Database Selection and Search Strategy

The following databases were selected because they collectively provide the broadest coverage of peer-reviewed business, management, and interdisciplinary research and are consistently recommended for systematic and scoping reviews in the social and management sciences (Gough & Thomas, 2016; Gusenbauer, 2024). Scopus and Web of Science provide comprehensive cross-disciplinary bibliographic indexing, while Wiley Online Library and Taylor & Francis Online ensure coverage of key management journals not fully

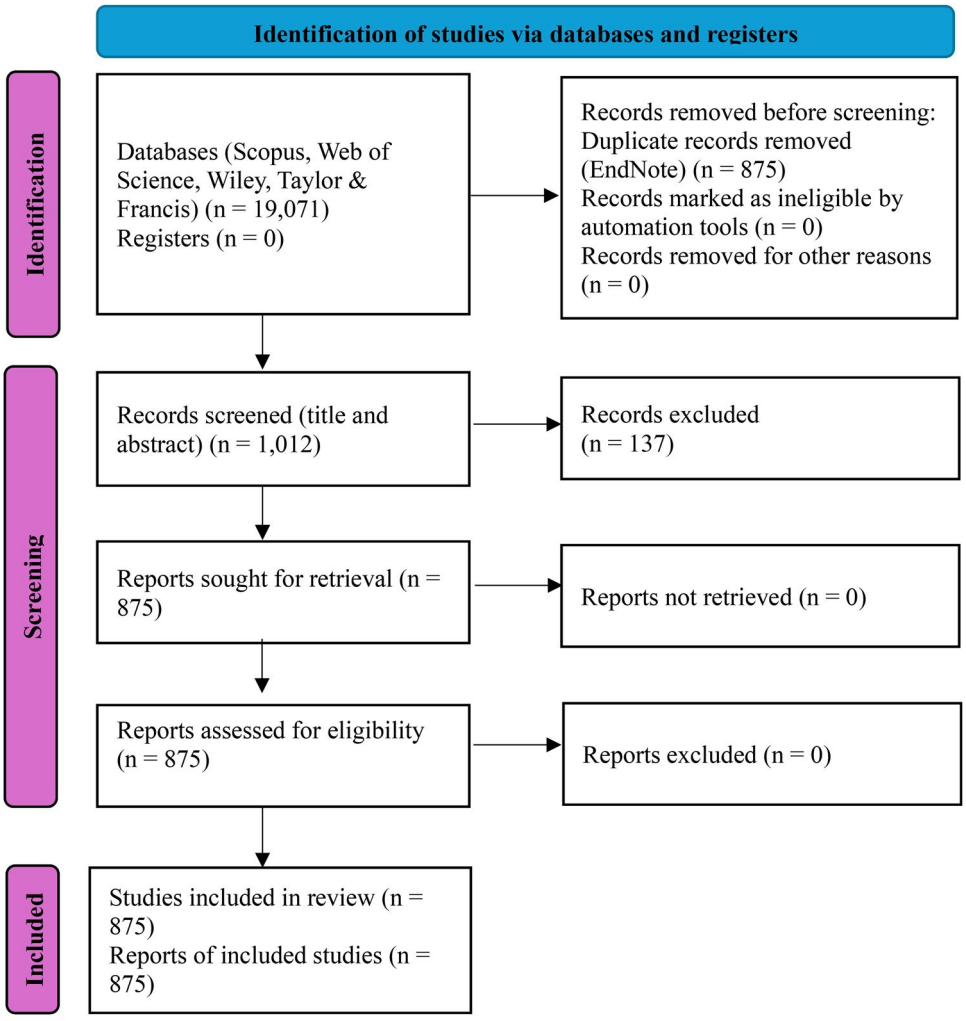


Figure 1. PRISMA flowchart. Authors' own elaboration. Drawn using Microsoft Word.

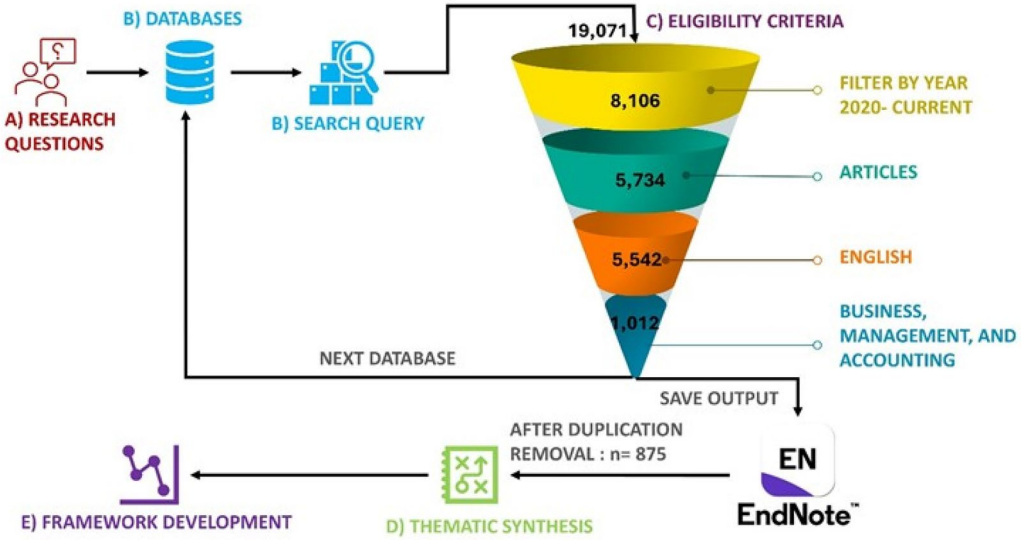


Figure 2. Methodological framework of the scoping review process. Authors' own elaboration. Drawn using Microsoft PowerPoint.

captured by the two primary databases. To capture the breadth of relevant literature, we applied the following query: ("AI agent*" OR "autonomous agent*" OR "agentic AI" OR "multi-agent" OR "intelligent agent*") AND ("decision making" OR "decision support" OR "autonomous decision*" OR "business decision*" OR "enterprise system*" OR "management decision*") AND (business OR enterprise OR organization* OR management OR strateg* OR "digital transformation"). The database search was conducted in June 2025.

C. Eligibility Criteria and Screening

Inclusion criteria limited records to journal articles that were (a) peer-reviewed, (b) in English, (c) published between 2020 and 2025, and (d) classified under business, management, and accounting subject areas. The 2020–2025 window was selected because agentic AI and large-scale autonomous decision-making systems emerged as distinct research streams following advances in deep reinforcement learning and large language models (LLMs) from 2020 onward, making this period the most relevant for capturing contemporary developments in the field. Titles and abstracts were screened independently by both authors, with any discrepancies resolved through discussion and consensus. Thematic coding was performed inductively, with codes reviewed iteratively by both authors until saturation was reached.

D. Dataset and Analysis Strategy

For each article, key metadata (title, abstract, keywords, year, and authors) were recorded, alongside thematic coding related to the six RQs. This allowed systematic mapping of areas of investigation, methods, and findings.

Following the screening, the 875 articles were retained as the primary dataset for mapping and quantitative analysis. Due to the breadth of the dataset, a smaller subset of studies was selected for in-depth qualitative synthesis and discussion, with the selection based on relevance, conceptual clarity, and representativeness of key themes identified during the analysis. This two-stage approach, combining broad dataset analysis with focused qualitative interpretation, is consistent with scoping review methodology and allows both comprehensive coverage and deeper insight.

E. Thematic Synthesis

Extracted data were synthesized qualitatively, with themes organized around the six RQs. Thematic clustering enabled us to identify converging insights and highlight gaps across domains.

F. Framework Development

Insights from the synthesis were consolidated into a taxonomy and conceptual framework, presented in later sections, to integrate disparate findings and guide future research.

The following RQs structured our review and analysis:

- **RQ1 Concept and Definitions:** How are *AI agents* and *autonomous decision-making systems* defined and conceptualized in the recent business literature? What characteristics (e.g., levels of autonomy, learning capabilities, role as decision support tools vs. fully autonomous actors) distinguish these agents?
- **RQ2 Application Domains:** In which business domains and functions have autonomous AI agents been applied? What are the primary use cases (e.g., finance, supply chain, marketing, HR, customer service), and which areas appear underrepresented in research?
- **RQ3 Techniques and Approaches:** What AI technologies and methodologies underpin these autonomous decision-making agents (e.g., multi-agent systems, reinforcement learning, machine learning models, expert systems)? How do these technical approaches differ by application context?
- RQ4 Outcomes and Impacts: What performance improvements or other benefits have businesses reported from using AI agents in decision-making? In contrast, what limitations or failures have been observed?
- **RQ5 Challenges and Risks:** What challenges, risks, or concerns are identified regarding AI autonomy in business decisions (e.g., ethical issues, lack of transparency, accountability gaps, integration with human teams, and regulatory compliance)? How are these challenges currently being addressed or mitigated?

- **RQ6 Frameworks and Future Directions:** What conceptual frameworks, taxonomies, or models have been proposed to understand or govern AI agents in business processes? What research gaps remain unfilled, and what future research directions are suggested to enable effective and responsible use of autonomous AI in organizations?

3. Results and discussion

The analysis is based on the dataset of 875 studies, which were used to identify patterns across application domains, levels of autonomy, and decision-making structures. For clarity and depth, representative studies are selectively discussed in the following sections rather than exhaustively listing all included articles.

Before addressing the six guiding RQs, an overview of the dataset is presented as Figure 3. The number of articles reveals a clear upward trend over time, increasing steadily from 69 in 2020 to 259 in 2025, reflecting the accelerating pace of research on AI agents and autonomous decision-making in This growth underscores the rising academic and practical interest in understanding how agentic AI systems influence managerial processes, organizational structures, and strategic outcomes.

Complementing this trend analysis, Figure 4 highlights the top 20 journals contributing to the dataset. International Journal of Production Research leads with 137 contributions, followed by IEEE Access (69) and International Journal of Computer Integrated Manufacturing (38).

Other prominent outlets include *Electronics*, *Expert Systems with Applications*, and *Journal of Simulation*, each with 22 articles, alongside management-oriented journals such as *Managerial and Decision Economics* (21) and *Technology Analysis & Strategic Management* (12). The distribution indicates that research is not confined to technical fields but spans operations and systems sciences as well as management and strategy, highlighting the interdisciplinary nature of the field.

Together, these descriptive patterns confirm that the literature is both rapidly expanding and widely dispersed across journals. This reinforces the importance of conducting a scoping review to integrate diverse insights and provide a structured synthesis. The following subsections address each of the six RQs in turn, presenting thematic findings and identifying conceptual and practical gaps.

These patterns suggest that the evolution of AI agents in business is not merely technological progression but a broader shift in how organizations conceptualize and distribute decision authority. The increasing presence of autonomous systems across functions indicates a transition from centralized human decision-making toward hybrid and distributed decision structures.

3.1. RQ1: Concept and definitions of AI agents in business

Across the 875 articles, a substantial subset of 613 studies explicitly mentioned definitions, conceptualizations, or autonomy of AI agents in their titles or abstracts. This subset was selected because it directly engages with the conceptual foundations of AI agents, making it essential for clarifying how agentic AI is understood in business contexts. Articles that focus solely on applications or performance outcomes,

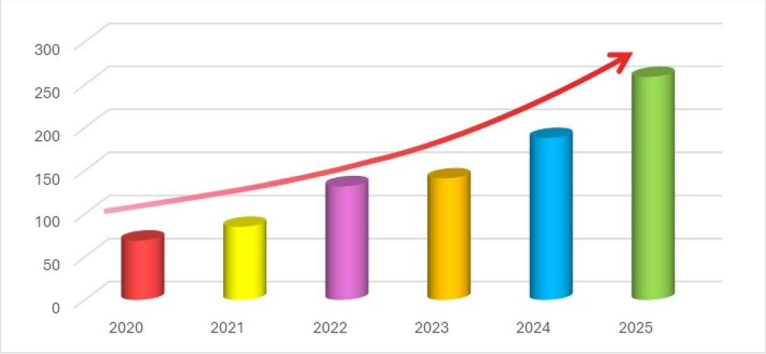


Figure 3. Annual distribution of reviewed articles (2020–2025). *Source:* Authors' own elaboration, PRISMA-identified dataset ($n = 875$). Drawn using Microsoft Excel.

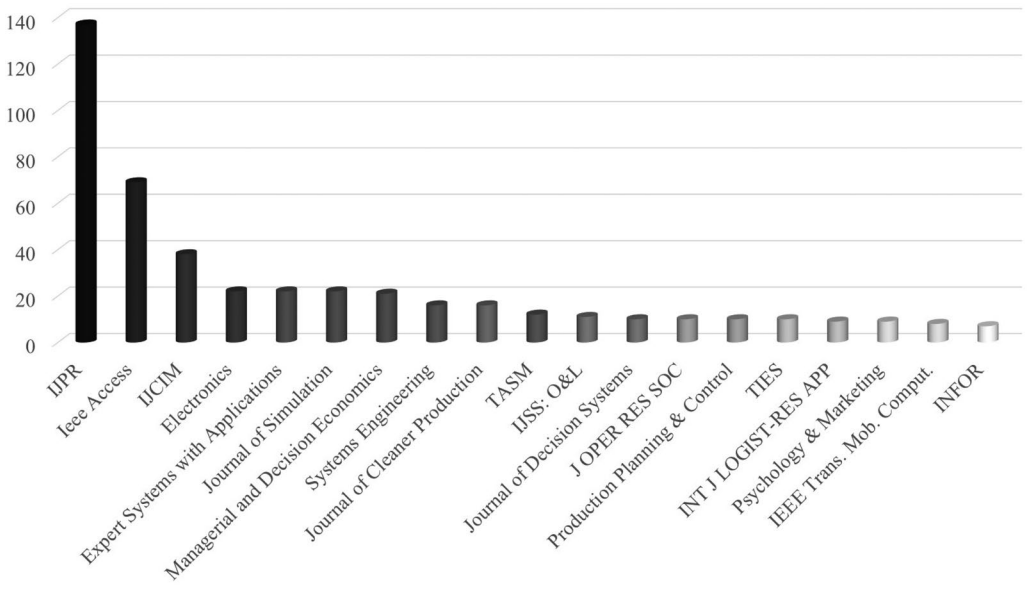


Figure 4. Top 20 journals contributing to the dataset. *Source:* Authors' own elaboration, PRISMA-identified dataset ($n = 875$). Drawn using Microsoft Excel.

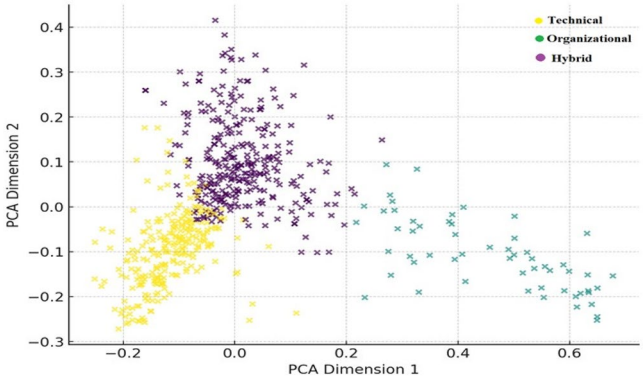


Figure 5. Clustering of AI agent definitions in business. *Source:* Authors' own elaboration, PRISMA-identified dataset ($n = 613$). Drawn using Python (scikit-learn, matplotlib).

without discussing definitions or autonomy, were excluded from this stage. This ensured that the synthesis was grounded in literature that most directly addresses questions of what AI agents are and how their roles in business decision-making should be understood.

To analyze this subset, we combined qualitative coding of abstracts with keyword-based clustering. Through inductive coding, three recurring definitional emphases emerged: a technical lens (agents as computational entities emphasizing autonomy and optimization), an organizational lens (agents as decision partners embedded in governance and managerial structures), and a hybrid lens (agents as socio-technical collaborators bridging algorithmic autonomy and organizational integration). These categories were corroborated through text clustering of abstracts using TF–IDF (term frequency–inverse document frequency) features and *k*-means clustering. The choice of $k = 3$ was determined through convergence between two independent approaches. First, inductive qualitative coding independently identified three recurring definitional emphases prior to computational analysis. Second, iterative testing of $k = 2$ through $k = 5$ was conducted, and $k = 3$ produced the most interpretively coherent and internally consistent clusters, with $k = 2$ collapsing meaningful distinctions between technical and hybrid perspectives, and $k = 4$ and $k = 5$ introducing redundant or fragmented subclusters without additional conceptual value. The three clusters had characteristic keyword patterns: Cluster 1 emphasized terms such as *model*, *systems*, *supply chain*; Cluster 2 highlighted *human*, *trust*, *consumers*; and Cluster 3 centered on *learning*, *reinforcement*, *control*. Figure 5 shows the clustering of AI agent definitions across 613 articles, where each point represents an article and the color corresponds to one of the three clusters.

This triangulation between qualitative coding and computational clustering strengthens the credibility of the three-cluster solution and the interpretation of three dominant definitional lenses in the literature.

Analysis of this subset reveals that definitions of AI agents in business literature are fragmented and discipline-dependent. Technical-oriented publications, such as those in *IEEE Access* and *International Journal of Production Research*, frequently describe agents as autonomous computational entities capable of perceiving environments, learning from data, and acting to achieve specified objectives. For example, Kazim et al. (2025) define swarm engineering systems as distributed AI agents whose autonomy enables self-organization in dynamic tasks. Similarly, Bi et al. (2024) define distributed decision-making agents as autonomous controllers that increase resilience in manufacturing systems. These definitions emphasize autonomy, adaptability, and algorithmic performance, reflecting their roots in computer science and operations research.

By contrast, contributions in business and management outlets stress organizational and managerial interpretations of agents. Alstete and Meyer (2020) conceptualize intelligent agents as decision-support partners embedded within knowledge management processes, underscoring their role in augmenting organizational memory rather than replacing human actors. Cannavacciuolo et al. (2024) likewise frame agent-based modeling as a way to conceptualize organizational dynamics, where agents are proxies for decision actors in socio-technical systems. These studies highlight the decision-making authority, governance implications, and human–AI collaboration aspects of agents, focusing on their integration into organizational structures rather than their technical underpinnings.

A third cluster bridges these perspectives through hybrid conceptualizations. For example, Gao et al. (2025) describe task-oriented agents in integrated networks as both algorithmic decision-makers and collaborative actors within broader organizational workflows. Zheng et al. (2025) present multi-agent coopetition strategies that combine computational autonomy with managerial decision rules, explicitly embedding AI into firm-level strategic interactions. Such works recognize that AI agents cannot be reduced to either technical systems or organizational actors but must be understood as socio-technical collaborators that embody both dimensions.

Taken together, these definitional clusters highlight a persistent lack of unified terminology in the literature. In technical journals, an *autonomous agent* often denotes a fully algorithmic system operating without human approval, while in management research, the same term may denote a co-leadership arrangement where humans retain oversight. This definitional inconsistency complicates cross-disciplinary dialogue and impedes the development of a coherent body of knowledge. To address this fragmentation, the review maps the literature along three dimensions: level of autonomy (decision support, semi-autonomous, fully autonomous), organizational role (operational efficiency, customer-facing, strategic decision-making), and conceptual lens (technical, organizational, hybrid). These dimensions are integrated into the taxonomy presented in Figure 6, which illustrates how existing definitions can be systematically organized.

A small subset of articles ("Unclassified" in Figure 6) did not provide sufficient conceptual detail in their abstracts to be reliably assigned to a specific autonomy level, role, or lens. These studies typically discuss AI agents in broad terms without clarifying the degree of autonomy or organizational role. While they reflect the diffuse usage of the term *agent* in the literature, they do not alter the overall patterns observed in the classified categories.

By synthesizing the 613 definition-related studies, this review clarifies the conceptual landscape of AI agents in business, enabling us to demonstrate how definitions vary across technical and managerial domains and propose a taxonomy that enables greater conceptual coherence. Such clarity is essential for both theory development and practice, helping practitioners to determine whether AI systems should act as decision-support tools, semi-autonomous collaborators, or fully autonomous decision-makers within business organizations.

This fragmentation in definitions highlights a deeper conceptual divide between technical and organizational perspectives, suggesting that future research should move toward unified socio-technical frameworks that explicitly link algorithmic autonomy with managerial decision structures. Critically, this divide is not merely terminological: operational autonomy, where AI agents execute bounded, repetitive, or optimization-driven tasks with minimal human intervention, has fundamentally different governance implications than strategic autonomy, where AI participation in high-stakes, ambiguous, or value-laden decisions raises substantially greater accountability and oversight concerns.

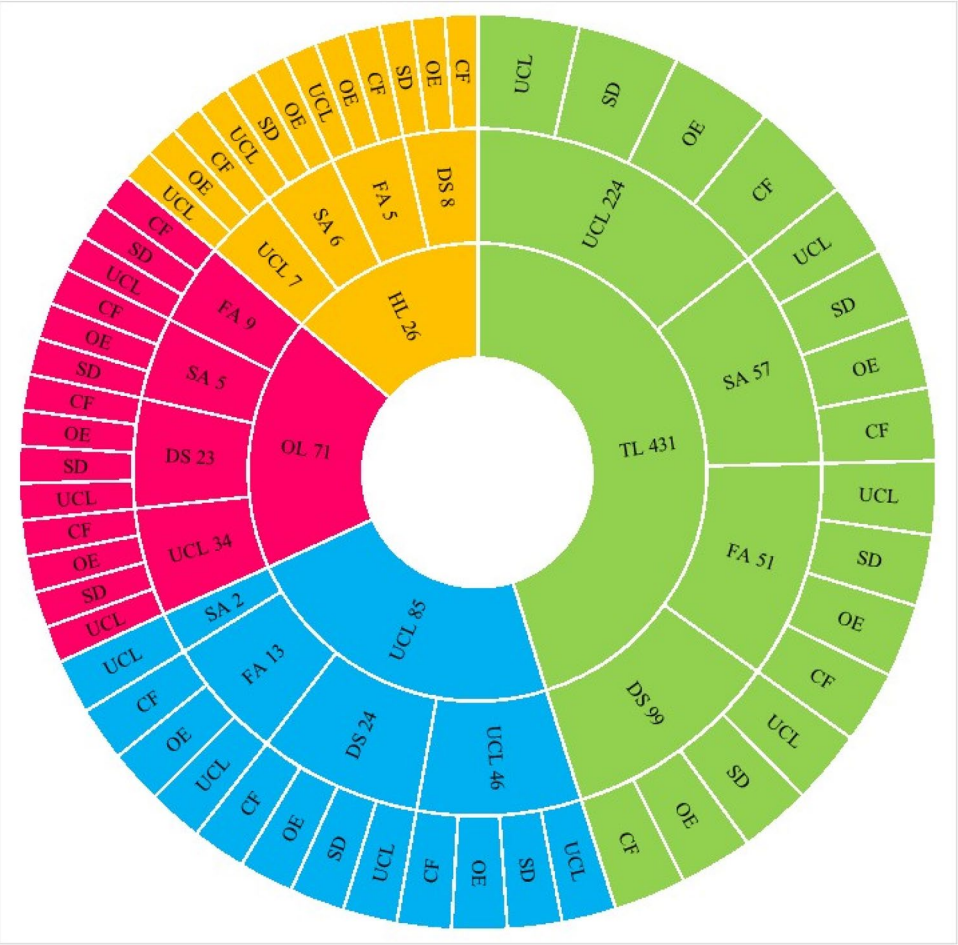


Figure 6. Taxonomy of AI agent conceptualizations in business. Authors' own elaboration. Drawn using Microsoft Excel. The taxonomy organizes conceptualizations by conceptual lens (Technical Lens = TL, Organizational Lens = OL, Hybrid Lens = HL, Unclassified = UCL), autonomy level (Decision-Support = DS, Semi-Autonomous = SA, Fully Autonomous = FA), and business role (Operational Efficiency = OE, Customer-Facing = CF, Strategic Decision = SD). Segment sizes correspond to the number of studies identified in the dataset ($n = 613$).

3.2. RQ2: Application domains and use cases

Across the 875 articles analyzed, AI agents have been applied in a wide range of business domains, though unevenly distributed. The largest share of studies ($n = 539$) situates AI agents within strategy and governance, reflecting scholarly attention to how autonomous decision-making reshapes organizational design, policy, and leadership structures (Cannavacciuolo et al., 2024; Hughes et al., 2025). This domain is followed by applications in technology and innovation management ($n = 405$) and human resources and workforce analytics ($n = 396$), where agents are framed as tools for talent optimization, organizational learning, and socio-technical integration (Safshekan et al., 2026; Stylos et al., 2025).

Operations and supply chain management ($n = 304$) represents a mid-sized but mature body of work, with MARL and distributed optimization widely studied for scheduling, logistics, and manufacturing resilience (Bi et al., 2024; Zheng et al., 2025). Applications in marketing and customer service ($n = 202$), while fewer in number, highlight the rise of intelligent chatbots, recommendation systems, and AI-driven customer relationship management platforms (Alstete & Meyer, 2020; Bonderud, 2025). Finally, finance and accounting ($n = 93$) are notably underrepresented, despite industry demand for AI agents in fraud detection, risk assessment, and financial forecasting.

This distribution suggests that current research prioritizes strategic governance and socio-technical integration over domain-specific or transactional functions. While technical advances in operations are well-studied, customer-facing and financial domains remain relatively neglected. As shown in Figure 7, this imbalance highlights clear opportunities for future exploration of AI agents in understudied business contexts.



Figure 7. Application domains of AI agents in business. *Source:* Authors' own elaboration, PRISMA-identified dataset ($n = 875$). Drawn using Microsoft Excel.

The observed concentration of research in certain domains and techniques indicates that current advances in AI-driven decision-making are unevenly distributed, with operational applications maturing faster than strategic and governance-oriented uses. This imbalance suggests a critical need for future research to address underexplored areas such as financial decision-making and governance frameworks.

3.3. RQ3: Techniques and technological approaches

Across the corpus, terms referring to techniques in titles and abstracts indicate a diversified but uneven methodological landscape. The most frequent families are simulation/discrete events (217 articles) and control/robotics (202), followed closely by reinforcement learning/multi-agent reinforcement learning (RL/MARL) (200) and optimization/operations research (157). These are complemented by agent-based modeling (ABM) (131), IoT/edge/cloud integration (75), evolutionary methods/metaheuristics (74), game theory/mechanism design (62), deep learning (59), natural language/conversational systems (53), forecasting/time series (23), and knowledge/rule-based approaches (20). This profile suggests that autonomous decision-making in business remains anchored in simulation, control, and operations, with learning-centric and language-centric approaches rising but not yet dominant.

3.3.1. Reinforcement learning/multi-agent reinforcement learning (RL/MARL)

RL appears extensively in operational decision contexts (e.g. allocation, routing, and scheduling) and increasingly for resource-aware autonomy. Recent examples include Yu et al. (2025), who discuss degradation-aware decision strategies in logistics using deep RL, and Younas et al. (2025), who propose a self-attention multi-agent RL architecture for coordinated decision-making. These studies emphasize adaptive policy learning under uncertainty, a hallmark of autonomous agents in dynamic business environments.

3.3.2. Agent-based modeling (ABM)

ABM is widely used to model distributed decision dynamics and emergent behavior. For example, Vidhya et al. (2025) simulate agent-driven orchestration for network functions in 5G contexts, while Tang et al. (2025) study cooperative behaviors in vehicle swarms via agent-based simulation. ABM papers in the dataset typically frame agents as entities making micro-level decisions, whose interactions yield macro-level business outcomes useful for exploring structure and policy design without requiring immediate deployment.

3.3.3. Evolutionary methods/metaheuristics

Population-based search methods remain common for hard combinatorial decisions under autonomy constraints (e.g. multi-feature decision problems and ethical or governance trade-offs). Representative works include Xie et al. (2025) on multi-feature decision problems using evolutionary optimization and

Wang et al. (2025) on governance decisions in commercial bribery contexts through metaheuristic modeling. These studies foreground solution quality and scalability rather than learning from interaction alone.

3.3.4. Deep learning

Deep neural models appear as perception/forecasting backbones inside agent pipelines (state representation, demand prediction, and anomaly detection) and less often as end-to-end decision-makers. While deep learning is not a plurality technique, it underpins many RL and optimization hybrids (e.g. function approximation for value/policy learning), as reflected across multiple journals in the corpus. For example, Ma et al. (2025) introduce a graph-based convolutional assignment neural network for heterogeneous multi-agent task allocation, achieving over 92% accuracy and reliability, showing how deep learning can improve coordination in complex agent systems.

3.3.5. Optimization and operations research (or)

Classic mixed-integer linear programming, stochastic programming, and heuristics remain foundational for bounded-autonomy decisions (e.g. constrained scheduling or routing). A typical pattern in the dataset is optimization wrapped with autonomy (e.g. agents call a solver under guardrails). For example, Zheng et al. (2025) propose dynamic flexible job-shop scheduling with autonomous agents interfacing an optimization core, Zhao et al. (2025) and Zhou et al. (2025a, 2025b, 2025c) couple centralized cooperative control with decision routines.

3.3.6. Game theory/mechanism design

Studies on strategic interaction and incentive alignment, critical when multiple agents or firms co-decide, are present in the dataset but smaller in number. Works in this stream formalize coordination, bargaining, and auctions, indicating growing attention to multi-stakeholder autonomy and governance. For example, Zhou et al. (2025a, 2025b, 2025c) formalize a differential game for autonomous cybersecurity defense, and Wang et al. (2025) integrate game theory with multi-agent RL for green job-shop scheduling, while Xu and Liu (2025) integrate game-theoretic models into procurement pricing under supply risk. Similarly, Shen et al. (2025) investigate collaborative incentive structures through a mechanism design lens, highlighting governance challenges in public–private partnerships. Sorooshian et al. (2026) further demonstrate this through a generative AI-assisted MCDM framework for supplier selection in the sustainable fashion industry, illustrating how AI agents can operationalize multi-criteria decisions in real business contexts. These cases illustrate how game theory provides a foundation for structuring multi-agent and multi-stakeholder decision autonomy in business contexts.

3.3.7. Control/robotics

Control-theoretical studies address safety, stability, and constraint handling, often prerequisites for granting autonomy. For example, Fereidooni et al. (2025) develop multi-agent deep-RL control to optimize urban traffic signal timing, illustrating how closed-loop control policies can coordinate distributed agents under real-time constraints. In a complementary safety vein, Ganeriwala et al. (2025) examine verification and certification pipelines for learning-enabled autonomous agents, showing how control-oriented assurance cases bound behavior within organizational tolerances. Broadening the lens, Ge et al. (2025) provide a survey of intelligent control and mapping techniques (e.g. robust/MPC/learning-based control) in relation to autonomy requirements such as stability margins and constraint satisfaction in dynamic environments. Together, these studies demonstrate how control logic structures agent behavior within enforceable limits that are governed by organizations.

3.3.8. Natural language/conversational agents

A relatively small number of studies focus on chatbots, dialogue systems, and LLM-augmented assistants in customer-facing and decision-support roles. They emphasize interaction quality, explainability, and human acceptance. For example, Yang et al. (2025) explore how LLMs can power conversational agents

in blockchain-enabled financial systems, underscoring their potential for both transparency and customer-facing applications.

3.3.9. Simulation/discrete events

Simulation is frequently used to evaluate policies safely before real-world rollout and remains an important means of testing agent decisions under realistic constraints (queueing, variability, and stochastic demand). Many of the 217 simulation-related papers combine simulation with RL/OR to validate performance envelopes prior to deployment. Recent examples include Zhang et al. (2025a, 2025b, 2025c), who use multi-agent deep RL with task decomposition for flexible manufacturing systems, and Zhang et al. (2025a, 2025b, 2025c), who explore traffic load–aware resource management strategies through multi-agent simulation in wireless networks. Similarly, Yu et al. (2025) apply simulation to develop battery degradation mitigation strategies in logistics systems. These cases illustrate how simulation not only supports operations in manufacturing but also extends to telecommunications and sustainable logistics, reinforcing its role as a key technique for stress-testing autonomous agent decisions before their deployment in business-critical environments.

3.3.10. Knowledge/rule-based approaches and forecasting

These streams appear in transparent, auditable agent rules and planning inputs. For example, Park and Shim (2025) explore rule-guided ground operations using interpretable expert systems. Similarly, Xi et al. (2025) examine demand information sharing and service investment in platform supply chains, while Tseng et al. (2025) apply deep RL to dynamic decision-making in medicine supply chains, as reported in *International Journal of Production Research*. These approaches stand out for their interpretability and regulatory compliance, qualities that remain essential in high-stakes autonomy where auditability is as important as accuracy.

3.3.11. Systems integration (IoT/edge/cloud)

A sizable subset addresses where decisions run (edge vs. cloud) and how agents coordinate across heterogeneous infrastructure. Wu and Guan (2025) explore multi-agent deep RL for cooperative edge networks, and Rodoshi et al. (2025) study multi-agent and centralized resource management in radio access networks. These works reveal architectural choices that affect the latency, reliability, and governance of autonomous systems (Babaei et al., 2025). The relative frequency of these techniques is summarized in Figure 8, which highlights the dominance of simulation, control, and RL, alongside the smaller but growing presence of natural-language and governance-aware approaches.

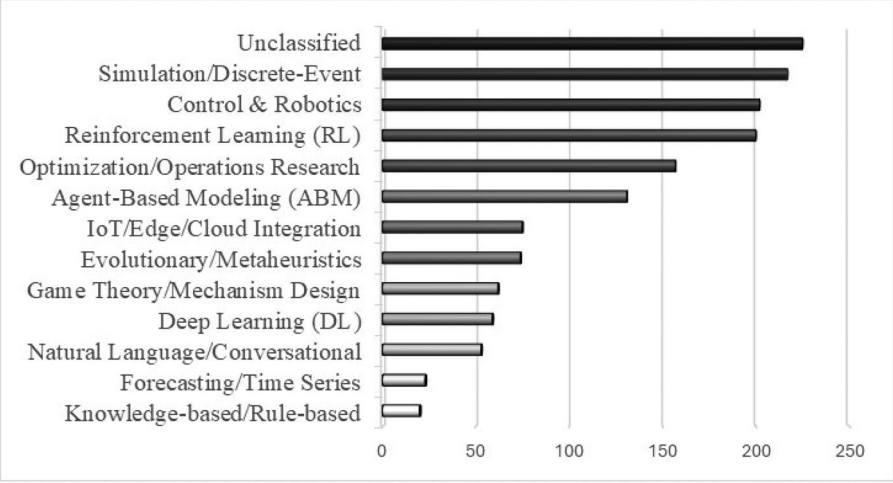


Figure 8. Techniques and technological approaches applied in AI agent research. Authors' own elaboration. PRISMA-identified dataset ($n = 875$). Drawn using Microsoft Excel.

Overall, the review highlights three notable findings. First, operations-centric techniques (simulation, control, RL, OR) dominate, consistent with the current prevalence of autonomy in operational/tactical decisions. Second, hybrid stacks are common: RL for policy learning, optimization for constraints, simulation for evaluation, and control for safety, reflecting layered autonomy aligned with business requirements. Third, language-centric and governance-aware techniques are growing but remain underrepresented, signaling opportunities for research on explainable, auditable, and dialogue-capable agents embedded in managerial workflows.

3.4. RQ4: Outcomes and impacts of AI agents

The most widely reported outcomes of AI agent adoption involve performance and efficiency gains. Many studies document improvements in productivity, better resource utilization, and reduced cycle time, confirming that efficiency is the main benefit of autonomy in business contexts (Zhou et al., 2025a, 2025b, 2025c; Zota et al., 2025).

Workforce and HR impacts also feature prominently, with research highlighting changes in managerial oversight, job roles, and employee trust when decisions are mediated by AI agents. This finding reflects growing discourse on the organizational and socio-technical implications of autonomy in daily business practices (Chen et al., 2024; Safshekan et al., 2026; Zercher et al., 2025). Innovation and strategic impacts are also reported, where agents are associated with accelerated innovation cycles, enhanced resilience, and support for competitive positioning (Alsharah, 2025; Dvouletý et al., 2025). These outcomes are expressed in terms of resilience and competitiveness, particularly in logistics and security contexts (Gupta et al., 2023; Tang et al., 2024). Finally, customer-facing outcomes, though less studied, point to improved service quality, satisfaction, and trust when transparency and explainability are ensured (Olan et al., 2025; Park, 2025; Wong & Wong, 2025).

As summarized in Figure 9, outcomes related to efficiency and workforce transformation dominate the current literature, while financial and customer-oriented impacts remain relatively underexplored, suggesting opportunities for future research.

A critical tension emerges from these findings: while technical literature consistently reports efficiency gains and performance improvements from AI agent adoption, management-oriented studies simultaneously identify growing deficits, particularly in safety, accountability, transparency, and human trust (Buzachis et al., 2020). This tension suggests that the measurable operational benefits of autonomy are being realized faster than the organizational and governance structures needed to ensure AI is used responsibly.

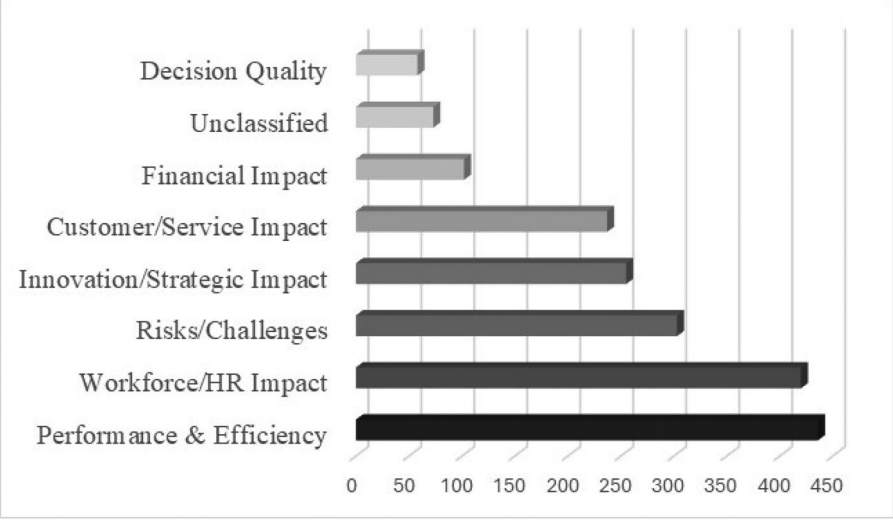


Figure 9. Reported outcomes and impacts of AI agents in business. Authors' own elaboration. PRISMA-identified dataset ($n = 875$). Drawn using Microsoft Excel.

3.5. RQ5: Challenges, risks, and concerns

The reviewed articles highlight multiple challenges associated with embedding AI agents into business decision-making processes. The most common concern is integration and technical complexity ($n = 204$). Several studies emphasize the infrastructural difficulties of aligning autonomous systems with legacy operations. For example, Zota et al. (2025) argue that developing agentic AI for IT operations (AIOps) systems requires reconciling heterogeneous data environments and existing business processes, while Zhou et al. (2025a, 2025b, 2025c) illustrate the high coordination complexity that emerges in distributed traffic control scenarios for autonomous vehicles. Complementary studies reinforce this view: Zheng et al. (2025) report reconfiguration challenges in flexible job-shop scheduling; Yi et al. (2025) underline the computational burden of deep RL for dynamic scheduling; and Teck et al. (2025) highlight the need for efficient heuristics in shop-floor integration.

Security and privacy risks ($n = 88$) are also prominent. Zhou et al. (2025a, 2025b, 2025c) stress that edge-intelligence architectures, although promising for distributed decision-making, are inherently vulnerable to advanced persistent threats. In parallel, Zota et al. (2025) caution that AIOps frameworks may exacerbate data-handling risks without robust safeguards. Karim et al. (2025) highlight the role of blockchain in enabling secure coordination among decentralized multi-agent systems, underscoring that distributed autonomy introduces new vulnerabilities if robust security mechanisms are not embedded.

Accountability and governance issues ($n = 77$) appear in works including Zhou et al. (2025a, 2025b, 2025c) and Yi et al. (2025), which underline the lack of clear responsibility when autonomous agents execute critical decisions. Workforce trust and acceptance ($n = 71$) remain contested. Zhou et al. (2025a, 2025b, 2025c) argue that opaque agent behaviors reduce operator trust, while Zhang et al. (2025a, 2025b, 2025c) show how agentic decision systems can influence consumer trust and acceptance of AI-mediated services. Finally, though less frequently addressed, ethical and bias issues ($n = 42$) and transparency and explainability ($n = 22$) remain important barriers. Younas et al. (2025) discuss fairness concerns in multi-agent coordination, while Yu et al. (2025) highlight the challenge of making agentic learning interpretable to stakeholders.

As shown in Figure 10, integration and technical complexity challenges dominate the literature, with more than 200 articles raising concerns about interoperability, scalability, and legacy system alignment.

Security and privacy, along with accountability and governance, form the second tier of concerns, reflecting the dual pressures of protecting data and clarifying responsibility for autonomous decisions. Workforce trust and acceptance are also significant, highlighting socio-technical tensions in adoption. Ethical issues and transparency remain underrepresented, suggesting a gap between technical development and responsible, explainable deployment.

Taken together, these results suggest that while efficiency and performance gains are well documented, the sustainability of agentic AI in business critically depends on addressing integration, governance, and trust challenges as well as developing ethical and transparent system designs.

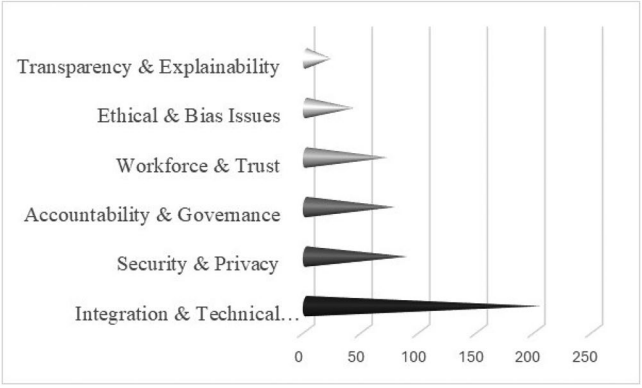


Figure 10. Challenges and risks associated with AI agents in business. Authors' own elaboration. PRISMA-identified dataset ($n = 875$). Drawn using Microsoft Excel.

A further tension is evident between the literature on operational autonomy and that on strategic decision-making. In operational contexts such as manufacturing, logistics, and scheduling, fully autonomous systems are increasingly accepted and validated. In contrast, studies addressing strategic and governance-oriented decisions consistently show a persistent demand for human oversight, reflecting the higher stakes, ambiguity, and accountability implications involved. This divergence signals that autonomy levels cannot be uniformly applied across business functions and must be calibrated to the nature and consequences of the decisions involved.

3.6. RQ6: Frameworks, taxonomies, and future research directions

The AAB integration model emerged inductively from the synthesis of findings across all six RQs. The five layers of the model correspond directly to the five thematic areas identified through the coding process: definitional lenses (RQ1), application domains (RQ2), technical architectures (RQ3), outcomes and challenges (RQ4 and RQ5), and governance frameworks (RQ6). This correspondence ensures that the model is grounded in the dataset rather than imposed a priori. During framework development, alternative structures were considered, including a two-layer model that separated technical and organizational dimensions and a four-layer model that separated outcomes from challenges. The two-layer structure was found to collapse important distinctions between application context and technical architecture, while the four-layer structure introduced redundancy without adding analytical clarity. The five-layer structure was retained, as it most parsimoniously captured the full analytical scope of the review. In terms of its theoretical contribution, the AAB integration model resolves the fragmentation between technical and organizational perspectives on agentic AI identified in RQ1 by providing a unified socio-technical structure that connects definitional clarity, application context, technical foundations, and governance considerations within a single coherent framework. Similarly, the autonomy taxonomy resolves the lack of standardized terminology around levels of AI decision authority, identified as a critical gap across the reviewed literature.

The synthesis of findings across the review indicates the need for more integrative conceptual tools to guide both research and practice. To address this need, we propose the AAB integration model, a layered socio-technical framework that connects definitional clarity, application contexts, technical foundations, and governance mechanisms. At its base, the model distinguishes three primary definitional lenses (technical, organizational, hybrid) that capture how scholars and practitioners conceptualize agentic AI. Building on this, it situates these conceptualizations within key application domains such as strategic management, operations, human resources, customer engagement, and finance. The next layer specifies the technical architectures that underpin autonomy, including RL, simulation, optimization, and governance-aware methods. Above this, the model integrates evidence of both positive outcomes, such as efficiency, innovation, and decision quality, and challenges related to bias, opacity, and accountability. At the top of the model is a governance and oversight layer, encompassing transparency, liability, and ethical oversight, reflecting broader calls for sustained human validation and oversight of AI-augmented decision frameworks, which, even in non-agentic settings, remain essential for managing bias, misplaced confidence, and contextual limitations (Sorooshian, 2025). By bringing together these dimensions into a coherent structure, the AAB integration model suggests that effective adoption requires alignment between technical sophistication, organizational embedding, and governance safeguards. This layered design is illustrated in Figure 11, which visualizes how agentic AI systems progress from conceptualization through application and technical implementation toward impacts and governance oversight.

In parallel, a new taxonomy of agentic autonomy can help differentiate the levels of decision authority granted to AI agents in business. This taxonomy classifies agents into three categories. Decision-support agents augment managerial judgments by offering recommendations or predictions; semi-autonomous agents share decision authority through collaborative or co-leadership arrangements, while fully autonomous agents execute operational or strategic choices with minimal human oversight. This taxonomy clarifies the continuum of autonomy and provides a useful lens for analyzing adoption decisions and their organizational implications. Specifically, operational autonomy refers to AI agents executing bounded, repetitive, or optimization-driven tasks such as scheduling, logistics, or resource allocation, where

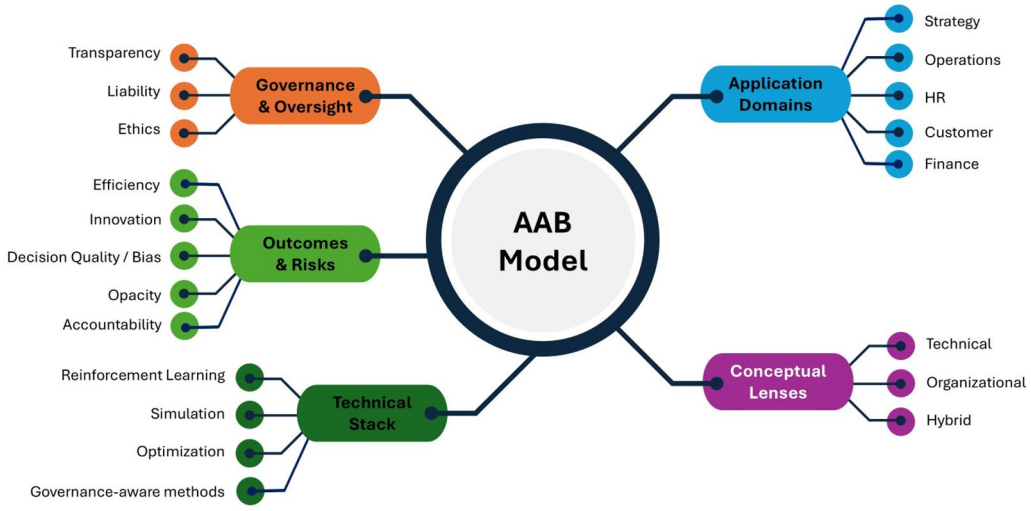


Figure 11. Agentic AI in Business (AAB) integration model. Authors' own elaboration. Drawn using Microsoft PowerPoint.

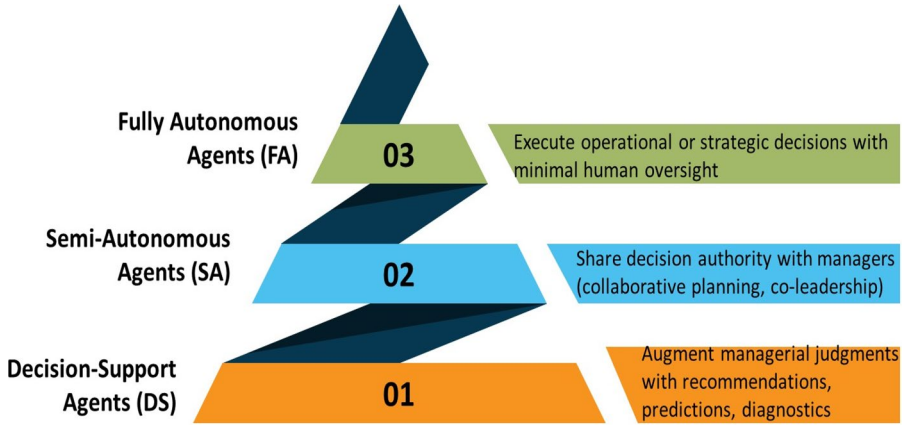


Figure 12. Taxonomy of agentic autonomy in business decision-making. Authors' own elaboration. Drawn using Microsoft PowerPoint.

performance is measurable and errors are recoverable. Strategic autonomy, by contrast, involves AI participation in high-stakes, ambiguous, or value-laden decisions such as strategic planning, workforce restructuring, or governance decisions, where accountability implications are substantially greater and human oversight remains essential.

The continuum of autonomy is represented in Figure 12, which classifies agentic AI into the three tiers and clarifies their organizational implications.

A summary of the proposed framework and taxonomy, along with suggested research directions, is presented in Table 2, which organizes these contributions into a structured reference for future investigations.

The findings of this review highlight several directions for future research. First, there is a need for greater conceptual clarity and standardized ontologies to reduce the definitional fragmentation surrounding agentic AI. Second, empirical studies should broaden their scope beyond operations and strategy to include underexplored domains such as finance and human resource management, where the organizational implications of autonomy are still poorly understood. Third, research should experiment with hybrid technical stacks that combine RL with explainable and fairness-oriented modules, thereby making autonomy both scalable and accountable. Fourth, there is a notable gap in longitudinal studies that assess the sustained effects of agentic AI adoption on business performance, innovation trajectories, and ethical compliance. Finally, future work should develop integrated governance frameworks that

Table 2. Proposed framework and taxonomy and future research directions.

Concept/proposal	Type (framework/taxonomy)	Description/contribution
Agentic AI in business (AAB) integration model	Framework	Layered socio-technical model linking definitions, applications, techniques, impacts, and governance
Levels of agentic autonomy (decision-support agents, semi-autonomous agents, fully autonomous agents)	Taxonomy	Classify agents into decision-support, semi-autonomous, and fully autonomous categories
Standardized ontology of agent roles	Research direction	Call for unified terminologies to reduce definitional ambiguity
Expanded application studies (HR, finance)	Research direction	Address underexplored domains beyond operations and strategy
Hybrid technical stacks with explainability	Research direction	Combine RL with fairness and transparency mechanisms
Longitudinal organizational studies	Research direction	Evaluate long-term impacts on performance, innovation, and ethics
Governance and accountability frameworks	Research direction	Develop mechanisms for trust calibration, liability, and ethical oversight
Unified socio-technical taxonomies	Research direction	Integrate technical, organizational, and governance layers into cohesive adoption models

Authors' own elaboration.

clarify accountability, calibrate human–AI trust, and balance efficiency with transparency. Together, these avenues suggest that progress in the field depends not only on technical advances but also on the articulation of socio-technical frameworks and taxonomies that enable the responsible scaling of autonomy in business contexts.

4. Conclusion

This scoping review has provided clarity on the definitions, applications, techniques, outcomes, and risks associated with AI agents in business decision-making. The findings demonstrate that while agentic AI enhances efficiency, agility, and innovation, it also raises critical concerns regarding trust, accountability, and transparency. We contribute two key advances: the agentic AI in business (AAB) integration model, which links conceptual, technical, and governance layers, and a taxonomy distinguishing decision-support, semi-autonomous, and fully autonomous agents. Together, these provide a foundation for guiding research and practice. Future work should expand studies into underexplored business functions, integrate explainability and fairness into technical architecture, and develop robust governance frameworks. By addressing these priorities, scholars and practitioners can better harness the potential of agentic AI while ensuring responsible and accountable adoption.

In terms of practical implications, this study's findings offer actionable insights for organizations seeking to implement AI-driven decision-making systems. By distinguishing between decision-support, semi-autonomous, and fully autonomous AI systems, organizations can better align technology adoption with their operational needs, risk tolerance, and governance structures. The proposed framework can support managers in identifying appropriate levels of autonomy and ensuring effective human–AI collaboration in decision processes. Organizations operating in high-stakes or heavily regulated environments are advised to begin with decision-support configurations, gradually expanding autonomy as governance mechanisms, transparency tools, and human oversight protocols are established and validated. Regarding policy implications, the increasing deployment of autonomous AI agents highlights the need for clear regulatory and governance frameworks. Policymakers should focus on establishing guidelines for accountability, transparency, and the ethical use of AI in decision-making. The findings suggest that different levels of AI autonomy require differentiated governance approaches to ensure responsible deployment while supporting innovation.

This study has several limitations. First, the review focuses on publications from 2020 to 2025, which may exclude earlier foundational research. Second, while a large dataset was used for mapping and analysis, only a representative subset of studies was discussed in detail. Third, the study relies on selected academic databases, which may not capture all relevant industry or gray literature. Future research should explore underrepresented domains such as financial decision-making and governance-focused applications of AI agents. In addition, empirical studies are needed to validate the proposed framework and examine how different levels of autonomy impact organizational performance and decision quality.

Further research could also investigate the integration of AI agents within hybrid human–AI decision systems across industries.

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Author contributions

CRediT: **Mostafa Ghane**: Data curation, Formal analysis, Methodology, Project administration, Resources, Validation, Writing – original draft, Writing – review & editing; **Shahryar Sorooshian**: Conceptualization, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Mei Choo Ang**: Validation, Writing – review & editing.

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Data availability statement

The data will be available per reasonable request to the corresponding author.

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